IBM CAPSTONE PROJECT

Jesna Salim

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OUTLINE

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EXECUTIVE SUMMARY

End-to-end Data Analytics project using real-world datasets.

Focus areas: data collection, preparation, analysis, and dashboard development.

Tools used: Python, SQL, Excel, Plotly, Dash.

EXECUTIVE SUMMARY

Data Collection

Retrieved job market data using APIs and HTTP requests.

Automated job count extraction by location and technology.

Stored structured datasets for analysis.

<u>Sub Points</u>

- JSON parsing using Python
- Reusable data collection functions
- Export to Excel for reporting

EXECUTIVE SUMMARY

Data Understanding & Preparation

Explored large survey dataset (65k+ records, 100+ features).

Identified data types and handled missing values.

Prepared categorical variables for analysis.

<u>Sub Points</u>

Dataset structure validation

Data cleaning and preprocessing

Encoding categorical features

EXECUTIVE SUMMARY

Feature Engineering

Created derived metrics (normalized compensation, grouped categories).

Transformed variables for better analysis.

Prepared datasets for visualization and insights.

EXECUTIVE SUMMARY

Exploratory Data Analysis (Key Insights)

Salary varies significantly by developer role.

Geographic differences strongly impact compensation.

Experience and job satisfaction influence salary levels.

<u>Sub Points</u>

Outliers detected in salary distribution

Experience vs satisfaction relationship

Employment type salary differences

EXECUTIVE SUMMARY

Visualization & Dashboard

Built interactive dashboards using Plotly and Dash.

Compared trends across time periods (recession vs non-recession).

Visualized sales, advertising spend, and vehicle categories.

<u>Sub Points</u>

Line charts for trends

Bar charts for comparisons

Pie charts for category contribution

INTRODUCTION

Purpose of the Report

- The purpose is to collect, prepare, analyze, and visualize real-world datasets to uncover meaningful insights related to developer job market trends and automobile sales performance.
- The report demonstrates how data-driven analysis supports informed business and strategic decision-making.

Target Audience

- Business stakeholders seeking insights for decision making.
- Hiring managers and recruiters interested in job market trends.
- Academic evaluators reviewing analytical methodology.
- Data professionals and learners exploring applied analytics workflows.

INTRODUCTION

Value of the Analysis

Market Understanding

- ❑ Provides visibility into developer job demand, skills trends, and geographic distribution.

Compensation Insights

- ❑ Identifies salary patterns across roles, experience levels, and employment types.

Business Performance Insights

- ❑ Evaluates automobile sales trends and advertising impact across economic periods.

INTRODUCTION

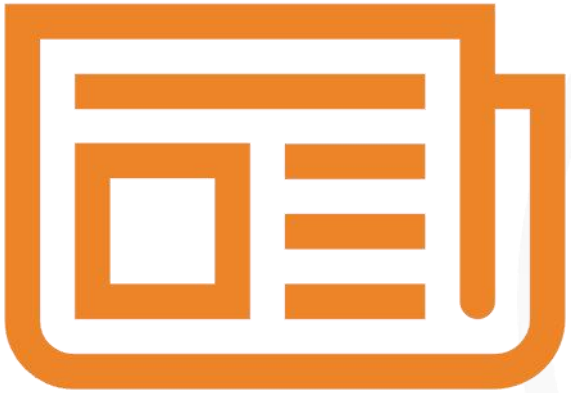
Decision Support

- ❑ Enables data-driven strategies for hiring, budgeting, and product positioning.

Sub Points

- ❑ Supports strategic workforce planning.
- ❑ Helps optimize marketing and advertising allocation.

METHODOLOGY



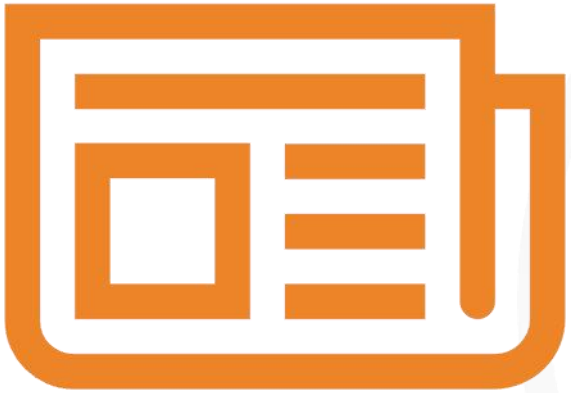
Data Sources

- ❖Developer survey dataset for demographic and compensation analysis.
- ❖Job posting API data to measure job demand by location and skills.
- ❖Automobile sales dataset for business trend analysis.
- ❖Supporting reference files for data enrichment.

Sub Points

- ❖CSV and JSON structured sources
- ❖Multi-source integration

METHODOLOGY



Data Collection

- ❖Retrieved API data using Python.
- ❖Imported datasets into pandas.
- ❖Automated job count extraction functions.
- ❖Stored datasets for reproducibility.

Sub Points

- ❖Programmatic extraction
- ❖Export to Excel

METHODOLOGY

Data Wrangling

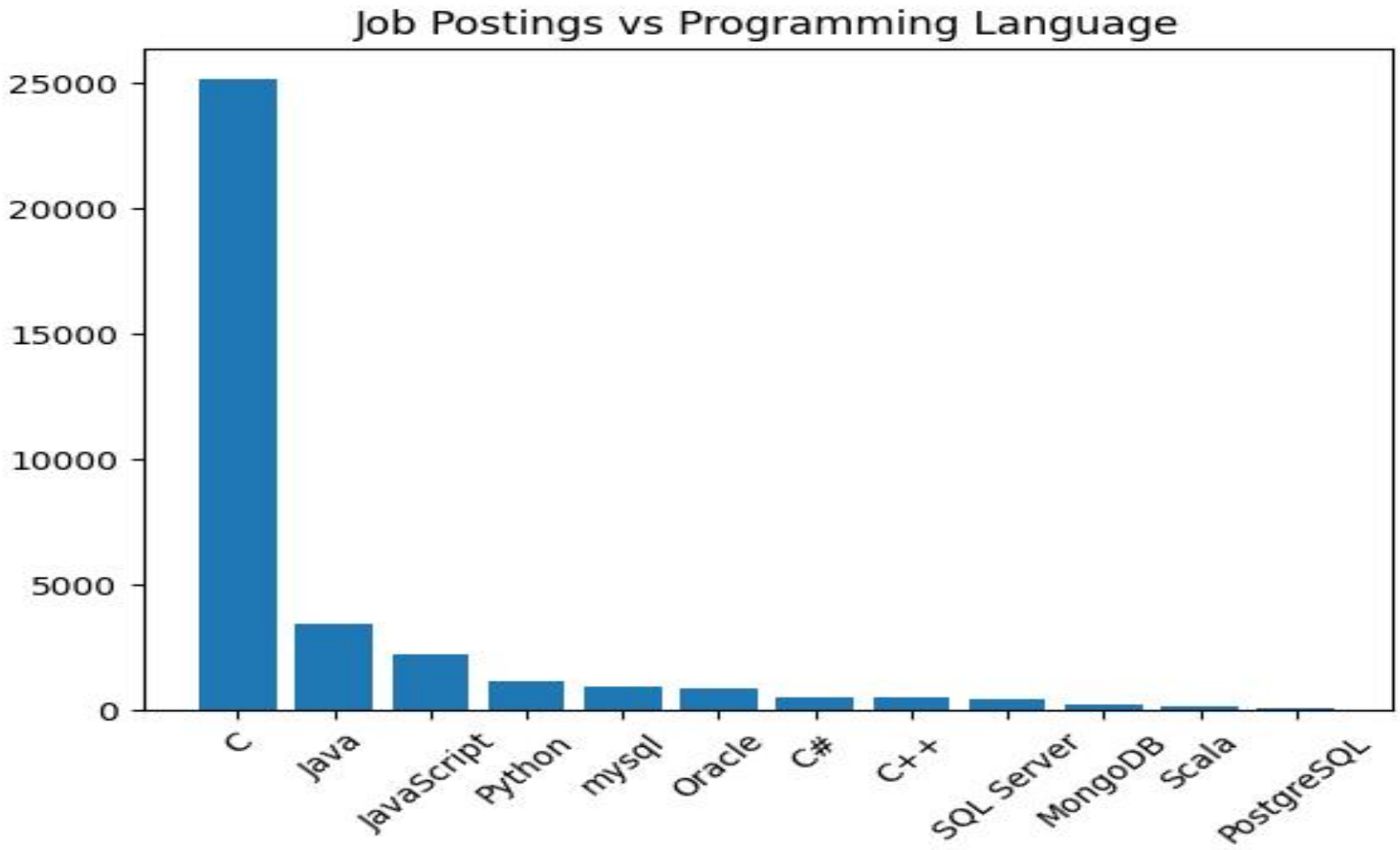
- ❖Cleaned missing and inconsistent data.
- ❖Transformed data types and created features.
- ❖Encoded categorical variables.
- ❖Validated data quality.

Sub Points

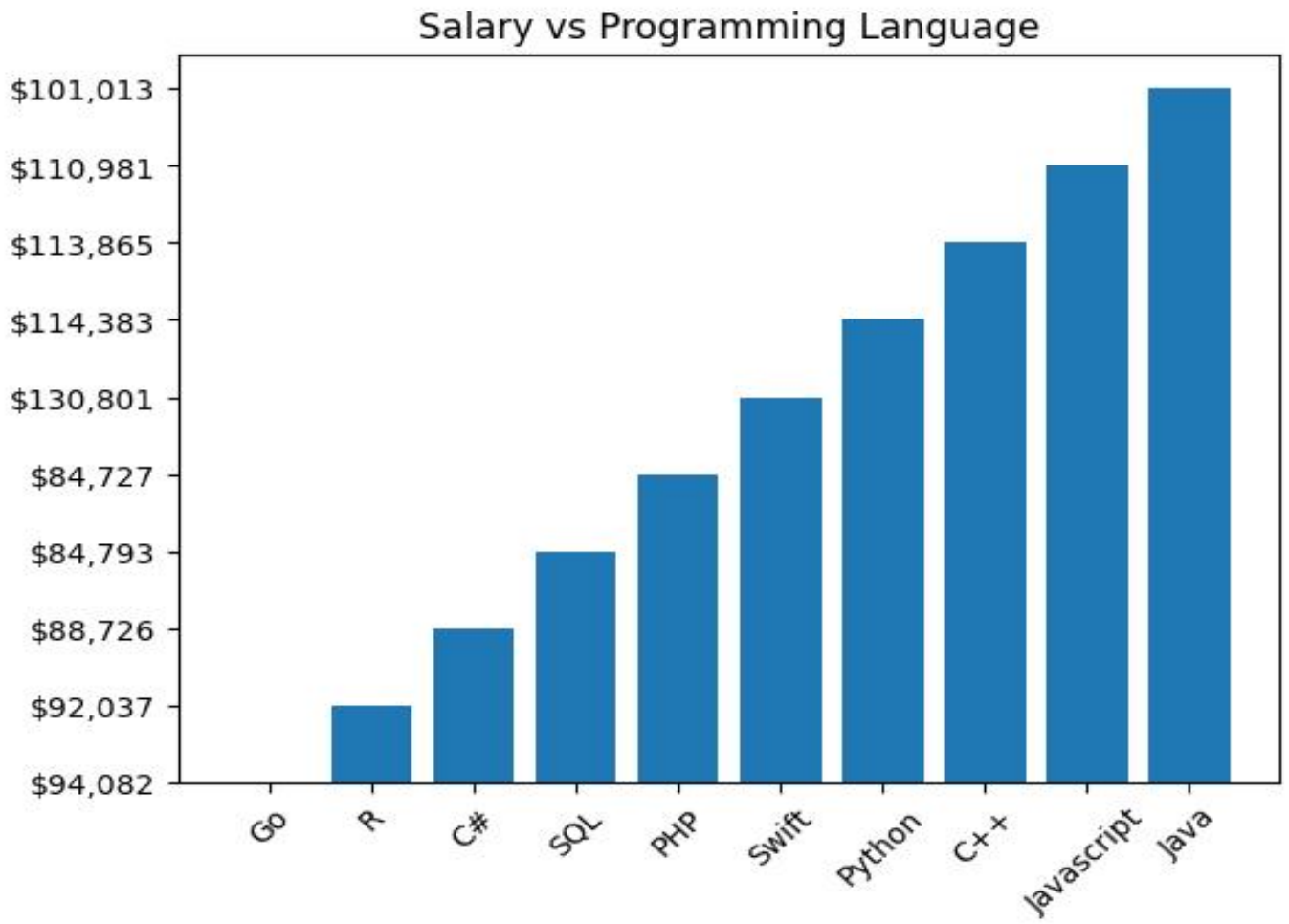
- ❖Outlier handling
- ❖Prepared analysis-ready dataset

Results

JOB POSTINGS

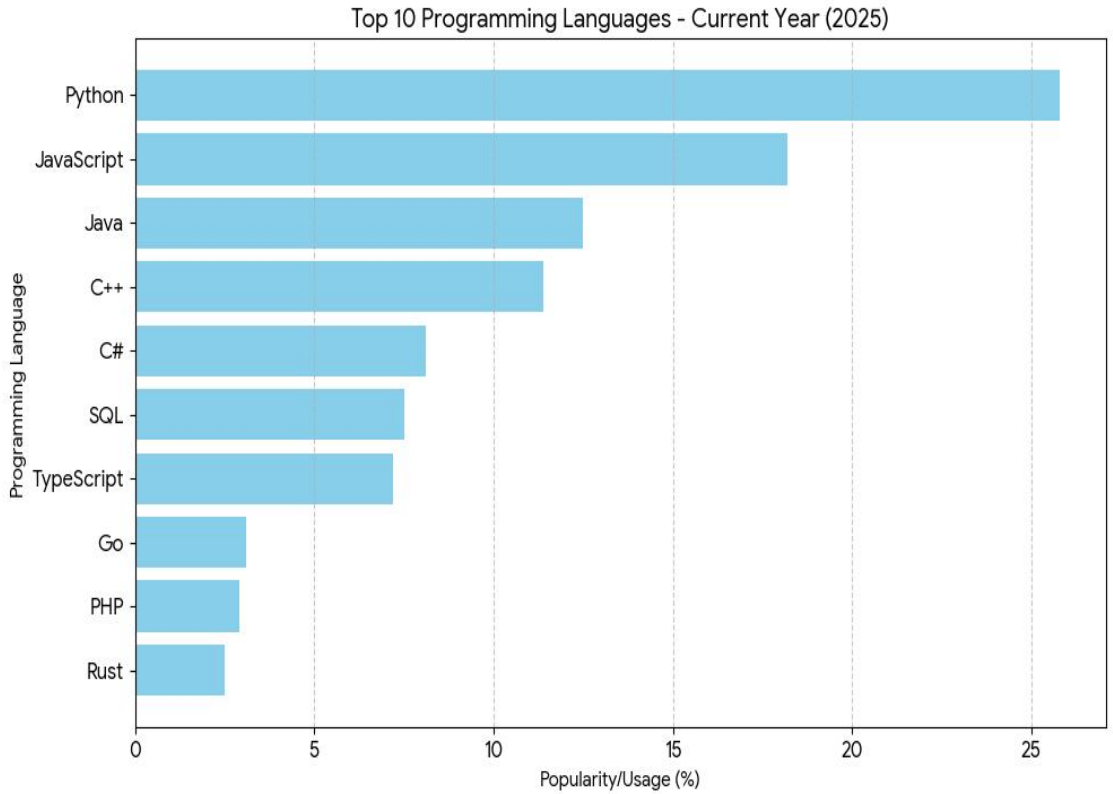


POPULAR LANGUAGES

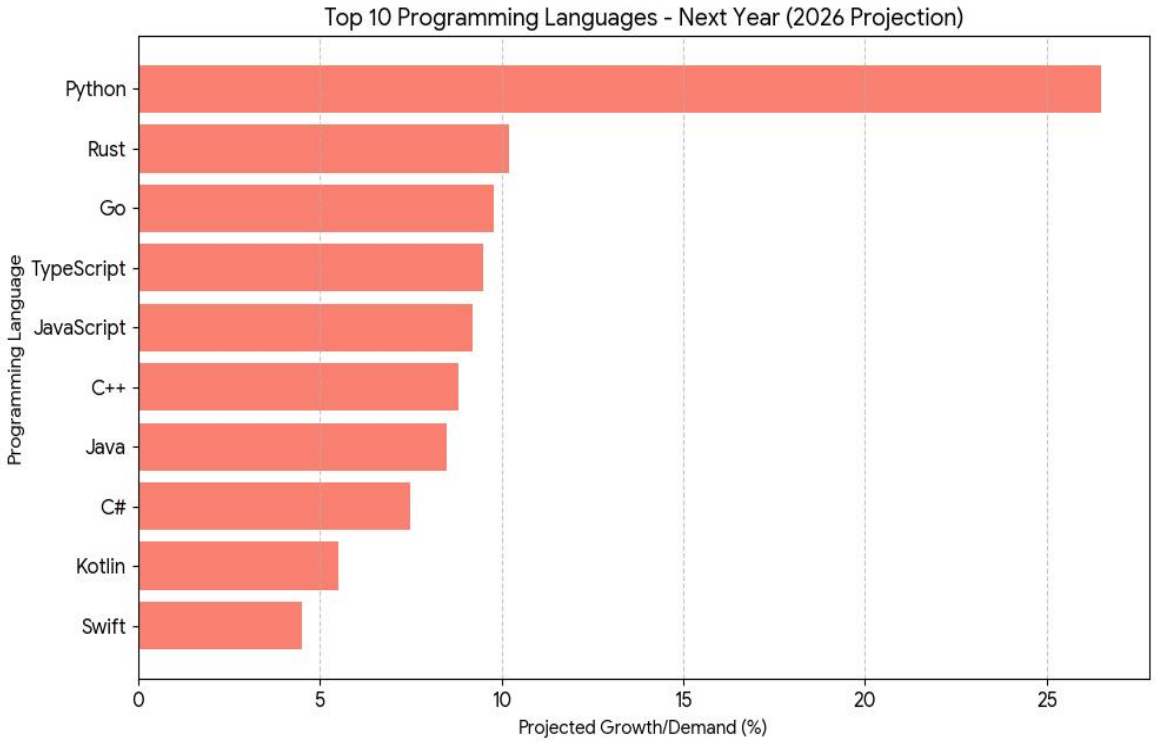


PROGRAMMING LANGUAGE TRENDS

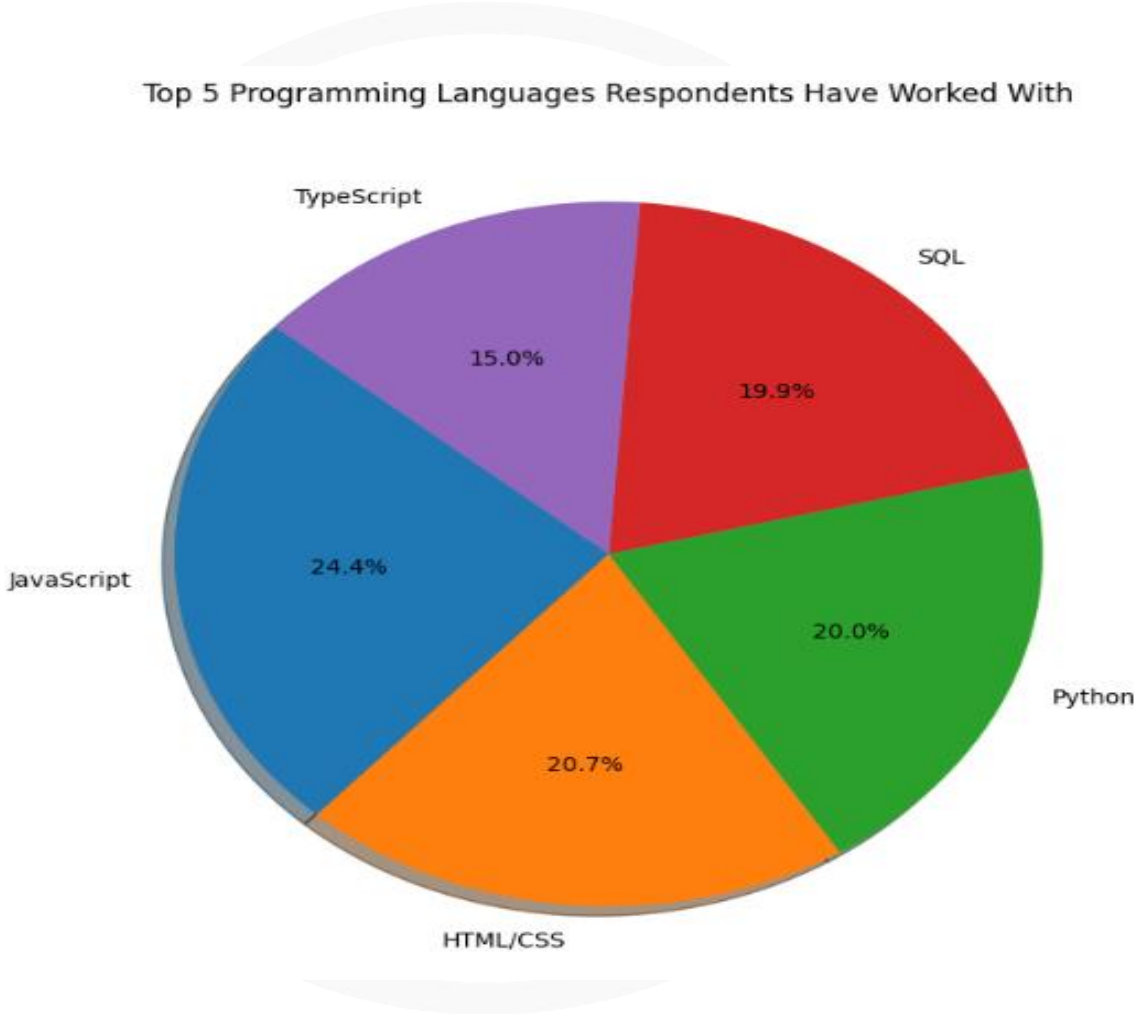
Current Year



Next Year

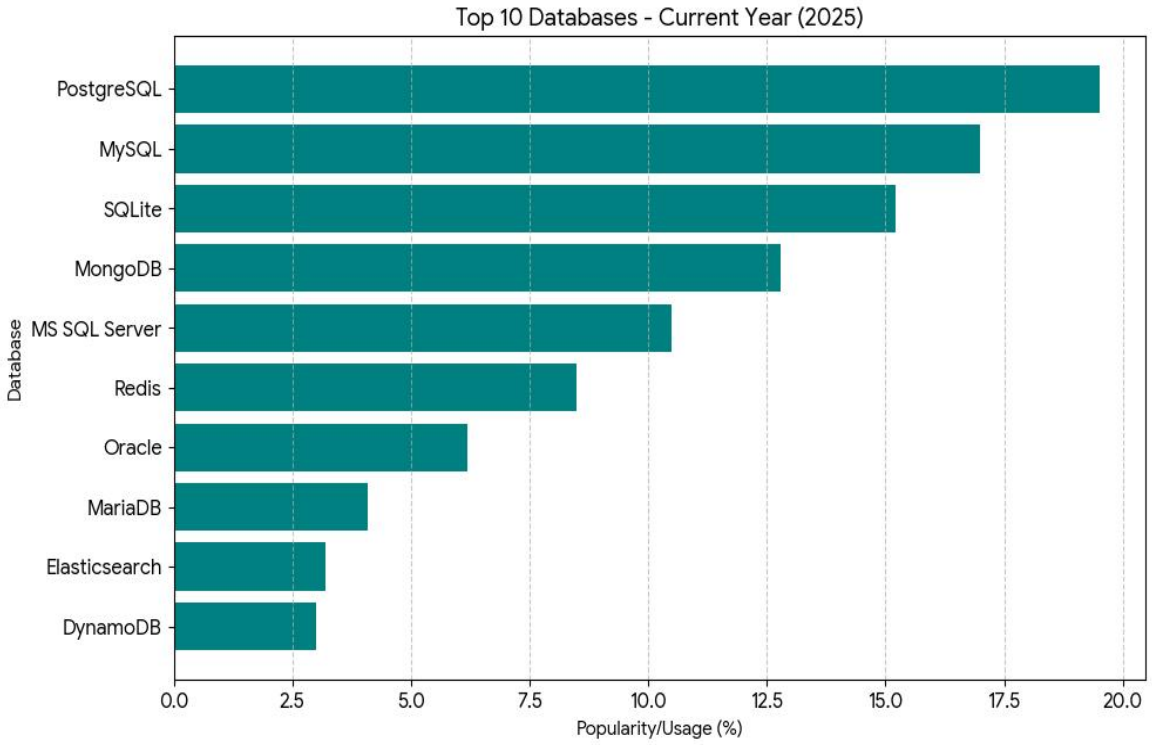


PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

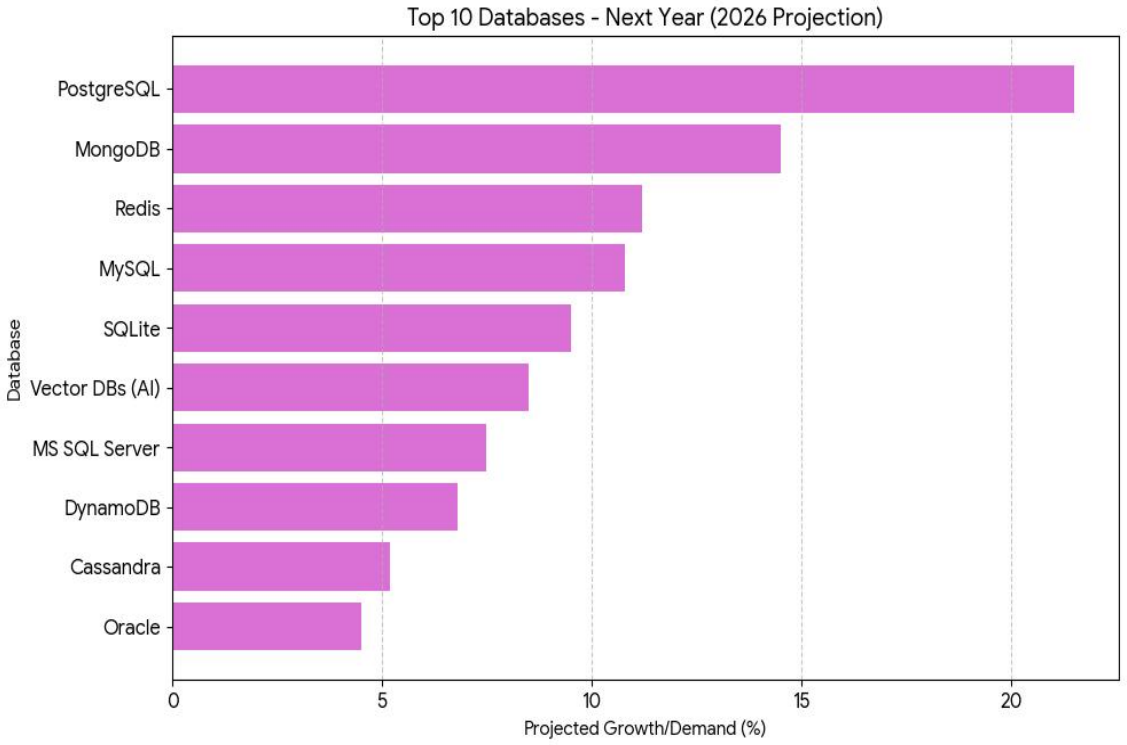


DATABASE TRENDS

Current Year



Next Year



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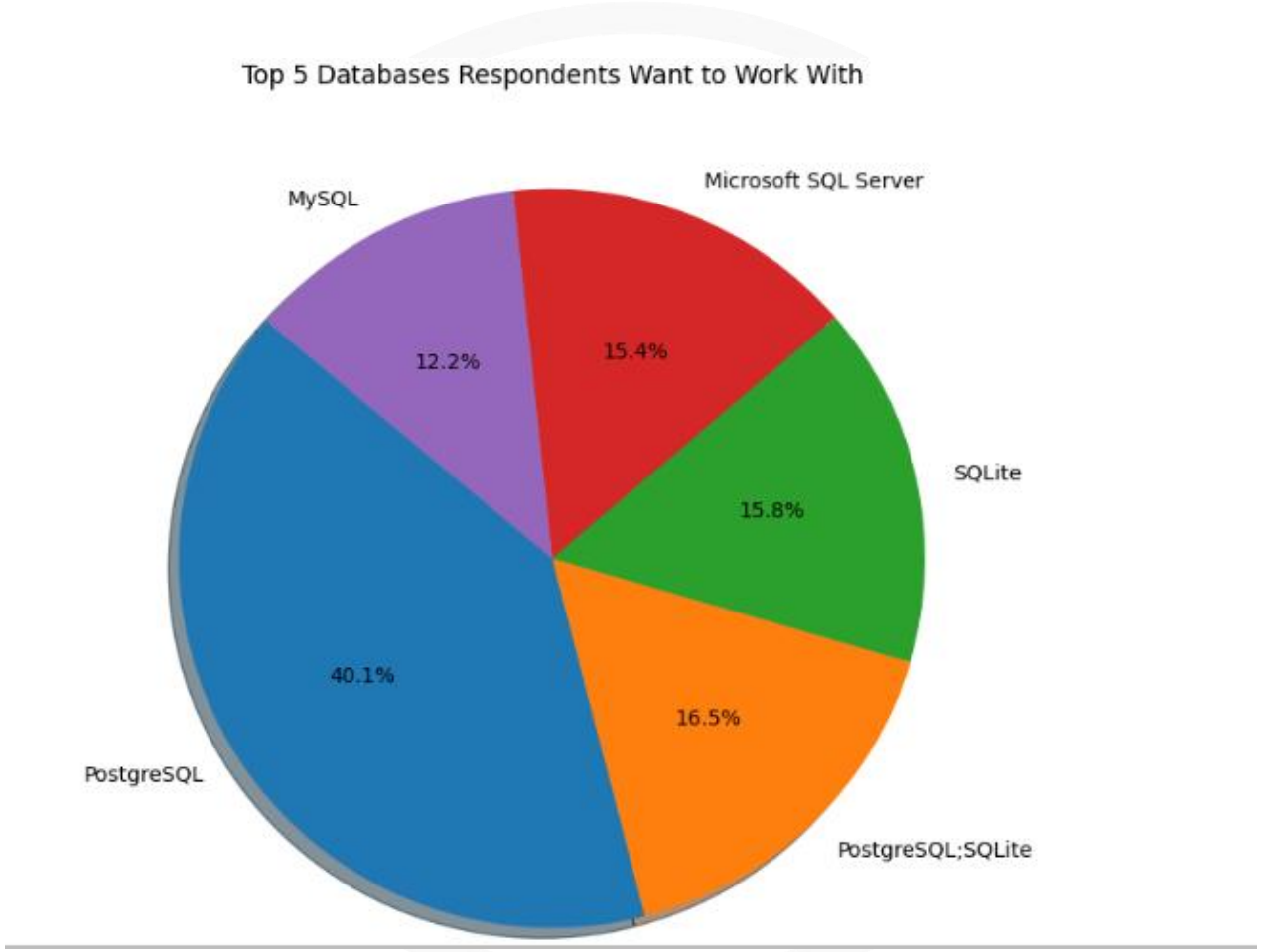
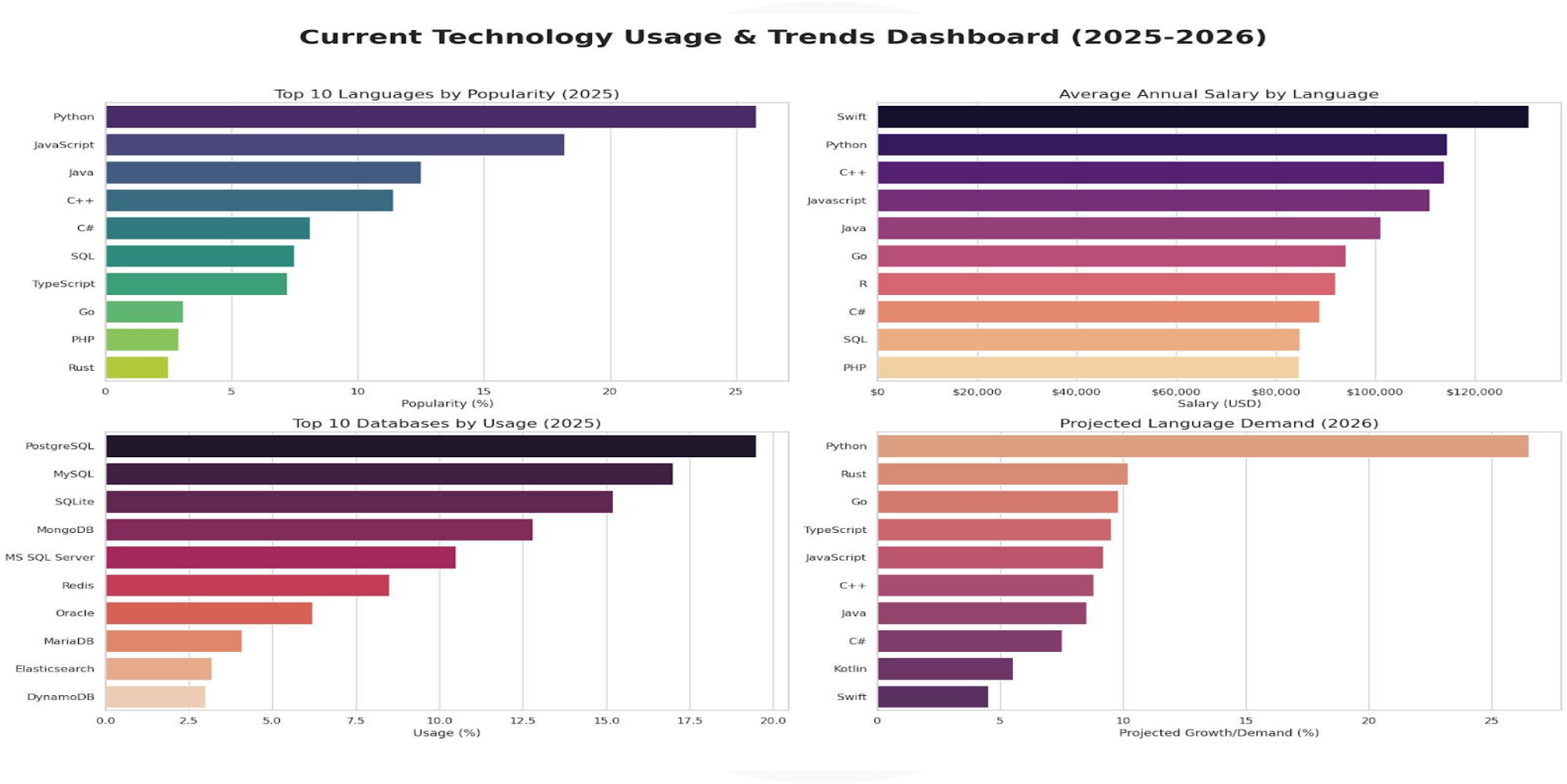
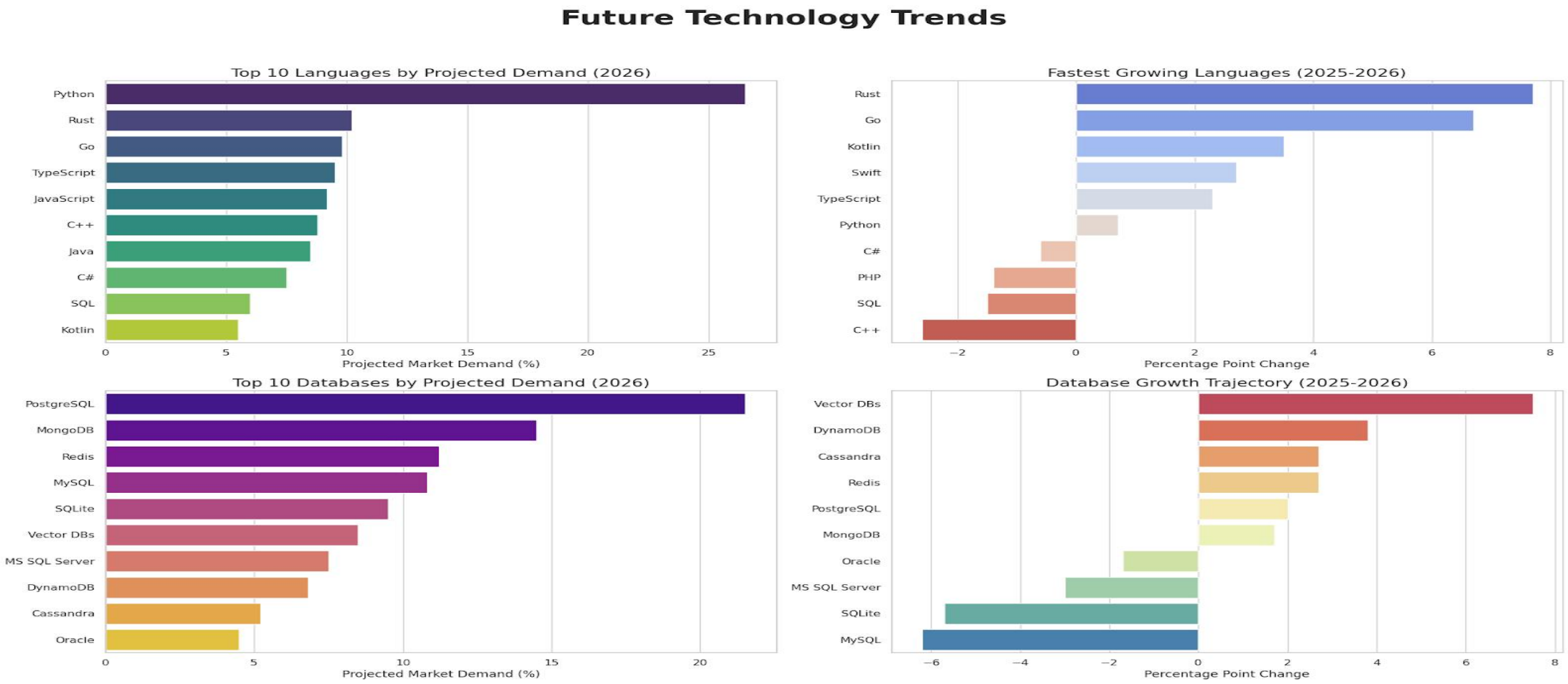


DASHBOARD

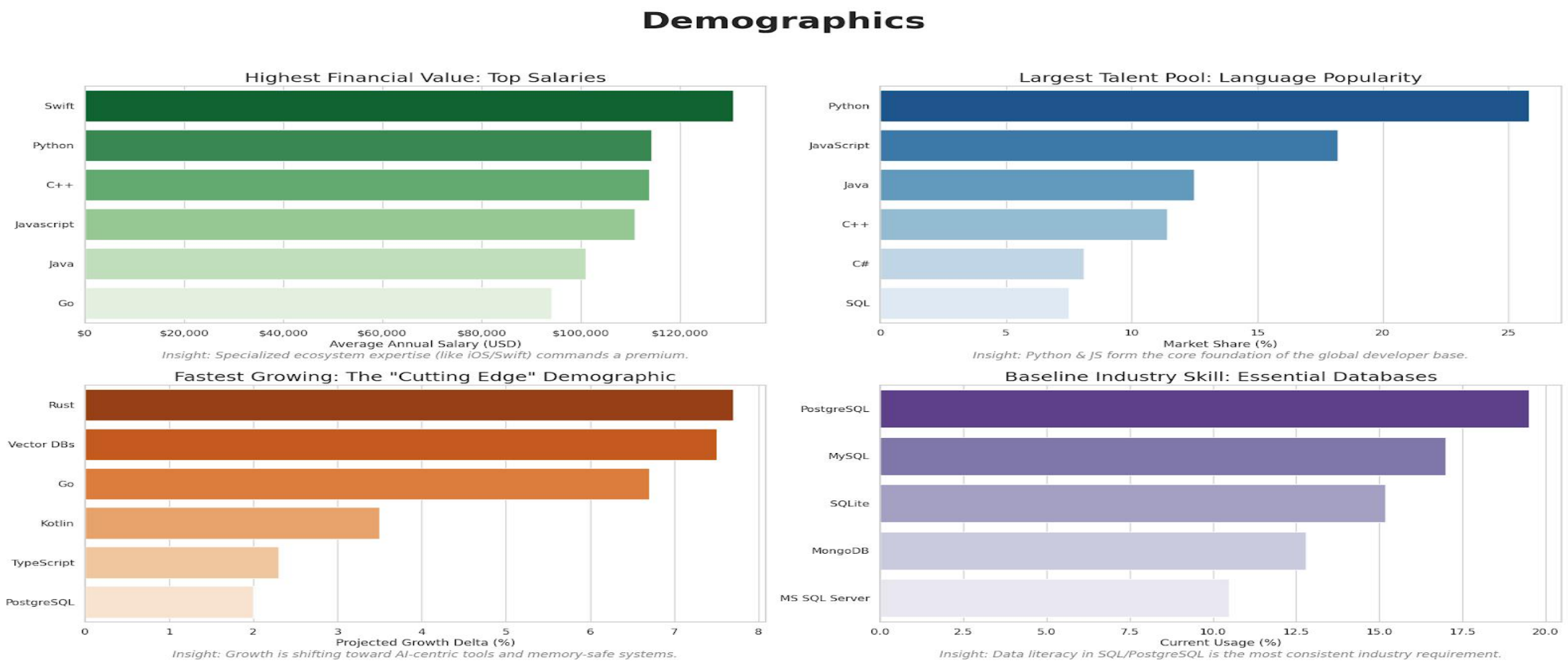
Current Technology Usage



Future Technology Trends



Demographics



Demographics

Metric	Leader / Peak	Insight
Highest Financial Value	Swift Developers	Specialized ecosystem expertise is highly rewarded.
Largest Talent Pool	Python & JavaScript	These languages form the "base" of the modern coding workforce.
Fastest Growing Skillset	Rust & Vector DBs	Represents the "cutting edge" demographic moving into AI and memory safety.
Baseline Industry Skill	SQL / PostgreSQL	Data literacy remains the most consistent requirement across all demographics.

DISCUSSION

OVERALL FINDINGS & IMPLICATIONS

Findings

- **Volume vs. Value Inversion**: There is a significant disconnect between job availability and compensation. C and Java dominate the market in total job volume (over 28,000 combined postings), while the highest salaries are found in specialized languages like **Swift ($130,801)** and **Python ($114,383)**, which have far fewer postings.
- **The "Workhorse" Standard**: Relational databases (**MySQL, Oracle**) and legacy languages (**C, Java**) remain the primary technical foundations of the industry. These "anchor" technologies provide the most consistent employment opportunities across the dataset.
- **Modern Stack Scarcity:** Modern technologies like **PostgreSQL** and **MongoDB** show relatively low job volumes in current postings compared to legacy counterparts, suggesting they are either highly specialized or currently underrepresented in large-scale corporate hiring relative to their popularity in the developer community.

Implications

- **Specialization Premium:** To maximize earning potential, professionals should transition from generalist roles to specialized ecosystems. The data implies that the "complexity premium" for AI (Python) and Mobile (Swift) allows for significantly higher salaries even in a more competitive or smaller job market.
- **Foundational Safety Net**: For those prioritizing job security, maintaining proficiency in **C** and **Java** is essential. These languages serve as a professional safety net, ensuring a high volume of available roles even if they do not offer the absolute peak salary.
- **The Data Literacy Requirement:** Database proficiency (SQL/Oracle) is not a standalone career path but a mandatory secondary skill. The consistent presence of these technologies across job postings implies that data management is a non-negotiable requirement for almost every high-level software engineering role.

CONCLUSION

- **Specialization Drives Compensation:** While legacy languages offer the most jobs, niche expertise in Swift and Python yields the highest financial returns, with salaries peaking between $114k and $131k.
- **Stability in Legacy Systems:** C and Java remain the industry's functional backbone, providing a combined safety net of nearly 29,000 job postings for those prioritizing consistent employment.
- **The Persistence of Relational Data:** Traditional databases like MySQL and Oracle continue to outperform modern NoSQL alternatives in job volume, cementing SQL proficiency as a mandatory "anchor skill."
- **Strategic Career Pathing:** The data confirms a clear trade-off: professionals must choose between the high-volume security of established enterprise stacks or the high-salary potential of modern, specialized ecosystems.

APPENDIX

- **Financial Value:** Swift and Python lead the market in individual earning potential, with Swift peaking at over $130,000.
- **Employment Stability:** Legacy systems and enterprise standards like C and Java remain the largest sources of job volume, offering widespread security for developers.
- **Future Trajectory:** Rust and Vector Databases are the primary growth drivers for 2026, signaling a shift toward high-performance systems and AI infrastructure.
- **Foundational Skills:** PostgreSQL and MySQL continue to be the essential database standards, providing the necessary data literacy for modern development roles.